- Places where the energy seems to shoot forth, and
- At where the force field seems to dissipate.

If you've been observing your fractal patterns carefully, then (and only then!) will the above make sense!

When we talk about energy shooting forth, it's in areas like the top left in Figure 3, and when we talk of the field dissipating, it's the circled part in Figure 4. Figure out for yourself where more such areas lie.

The thing is *infinitely* zoomable, only limited by computing horsepower! So when we say "find out such areas," *don't* limit yourself to the main fractal—zoom in to totally random areas, zoom in even more, find more and more balls and spirals and such, and try to find the "energy" and the "dissipation"!

Note, though, that you'll have to set high values for the number of iterations as you zoom in further, or you won't get the real picture.

An aside: talking about computing horsepower, back in 1995, on our 386 machines, the screens that now get generated for you in a few seconds would take hours, if not days!

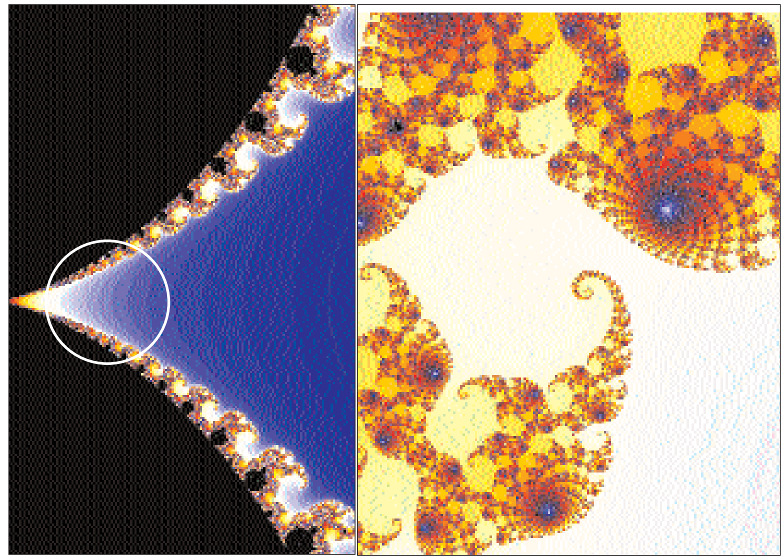

Figure 4: If you zoom into the areas of what looks like an unmentionable body part, you'll find the landscape is vastly different. Softer, gentler curves. No shooting stars

Natural Fractals

We did say that fractals showcase nature. How...? The fact is, nature abounds in fractal formations. Take the Java applet at **http://snipurl.com/fract1 (http://math.rice.edu/~lanius/frac/koch/koch.html**). There, the second iteration onwards, you'll see what looks like a snowflake. As you keep iterating, you'll find it getting more and more complex... and it becomes self-similar. That is, everywhere along the edges, it looks like the main shape itself. There are only so many iterations you can do with that particular applet, but you'll get the idea.

Coastlines, too, are something like this. If you were to measure a coastline with a mile-long ruler, you'd get a certain length.

Make it a foot-long ruler, and you'll see that it's longer than you thought. And so on. Ultimately, you'll find you just can't measure a coastline—it's infinitely long!

That's where the fractal part comes in—coastlines are *infinitely* detailed.

Mountains, snowflakes, leaves... are all fractals. Look at a leaf: it's got a main line with "branches." These branches have similar branches, and so on—and ultimately, if you examined it under a microscope, you'd probably find the same pattern repeated infinitely.

The above is just a rough introduction, and fractals in nature is an immensely interesting subject.

Visit **http://snipurl.com/fract2 (http://kluge.in-chemnitz.de/documents/fractal/node2.html)**, and **http://snipurl.com/fract3 (www.miqel.com/fractals_math_patterns/visual-math-natural-fractals.html)**.

How It Happens

Someone once said the sales of a book would drop by a factor of two for each equation inserted. In that vein, we'll explain how fractals work without using a single equation! However, you'll *need* to know about imaginary numbers, otherwise you might as well skip this. OK, to ease up a bit, think of the square root of -1. It cannot *have* a real square root, of course, because no real number, when squared, will give a *negative* number! The square root of -1 is therefore called "i", standing for "imaginary"; it's a number that's not on the line of real numbers.

Now, think of a number *plane*. "2 + 3i" represents the point which is 2 units from the origin on the X axis, and 3 units from the origin on the Y axis. See how the concept of an imaginary number leads to a number plane instead of just a number line?

And now, take that number and square it. You get (-5 + 12i), because i^2 is -1. Square it again. You get (-119 - 120i)... and so on. If the number runs away to infinity, it's not "part of the fractal set," and is painted white. If the number does *not* go to infinity, and stays within limits, it's part of the set, and is painted black. Take (0,0): it's obviously part of the set, because no matter how much squaring you do, it'll remain 0.

So how do you get colours? A point is marked with a certain colour depending on how *long* it takes to move to infinity. Light colours are for those that become huge fairly quickly; dark colours are for the ones that take longer to become large. But how many times does one keep squaring? Obviously, one sets a stop point: "we'll iterate 200 times," for example. After the 200 iterations are done, a colour is assigned, based on how far it has moved to infinity (or how close it has stayed to zero).

You'll often see the word "Julia" in fractals. This is the "complement" of the main set: there is a Julia set for each point in the plane. A point *within* the main set will generate a Julia set with only one shape; one outside the set will result in two shapes close to each other.

Getting Down To It

Now, it's rather pointless telling you what parameters to change and why and how. FF features a comprehensive tutorial. We'll tell you this, though—that seemingly unrelated parameters, when changed at the same time, can lead to a drastically different picture.

Also, just try changing the real and imaginary parts of the "perturbations" in FF a wee bit and be amazed! In other words, just keep playing around...

It's a matter of discovery, and we can't teach you fractal art. Just explore *as many regions of the initial fractal* as you can.

If you come across something truly unique—such as interesting symmetries—do remember to save them and send them in to us.

Welcome to psychedelia—the natural way! ■

ram_mohan@thinkdigit.com